Separating Uncertainty from First-Moment Shocks: Evidence from BRAC

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Abstract

Identifying the causal effects of uncertainty is challenging because uncertainty shocks typically coincide with first-moment shocks to economic activity. We exploit the 2005 U.S. Base Realignment and Closure (BRAC) process as a natural experiment that cleanly separates uncertainty from subsequent economic changes. During the BRAC process, communities face several months of uncertainty regarding future military base closures and realignments, while the resulting employment changes are not implemented until years later. This setting allows us to isolate the effects of a pure second-moment uncertainty shock. Using local attention to BRAC as an instrument for uncertainty, we find that a one-standard-deviation increase in uncertainty reduces employment and labor force growth by about 1 percentage point each, while having little effect on the unemployment rate. We also find a 1.4 percentage point decline in the share of new construction loans, consistent with weaker local economic activity. These results provide causal evidence that uncertainty depresses economic activity even in the absence of contemporaneous first-moment shocks.

JEL Classification: E32, J21, R23, H56

Keywords: uncertainty shocks, natural experiment, BRAC, local labor markets, labor force participation, housing construction

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Acknowledgments: We thank Francesco Bianchi, Nick Bloom, Steve Davis, Matteo Iacoviello, Zheng Liu, Juan Londono, Svetlana Makarova and participants at the SITE conference on "*The Macroeconomics of Uncertainty and Volatility*", the 5th Biennial Conference on "*Uncertainty, Economic Activity, and Forecasting in a Changing Environment*" and Southern Economic Association Annual Meeting for many helpful comments. Special thanks to the Becker Friedman Institute for Economics Data Acquisition Grant program for access to CoreLogic data, Scott Baker for his help with online news data, and Carrie Mead with the US Army for additional context on the BRAC process.

1 Introduction

Uncertainty is an important source of macroeconomic volatility, and has often been modeled as second-moment shocks (Bloom, 2009; Gourio, 2012; Leduc and Liu, 2016; Basu and Bundick, 2017; Fernández-Villaverde and Guerrón-Quintana, 2020; Dietrich et al., 2022). However, crashes in levels of production or demand tend to occur in uncertain times, and uncertainty shocks can cause recessions. This endogeneity makes separating second-moment shocks from first-moment (level) shocks difficult in practice, which in turn makes isolating the impact of a pure second-moment uncertainty shock challenging.

To identify the effects of a pure uncertainty shock, this paper exploits a natural experiment involving domestic U.S. military base closures, which neatly separates any possible first-moment shocks from the uncertainty shock. The uncertainty shock originates with an initial proposal for military base closures and personnel changes, developed through a strictly internal process at the U.S. Department of Defense (DoD). This proposal is then followed by several months of public debate and revision before the final changes are announced, generating substantial uncertainty in locations with military bases. This setting also alleviates concerns about reverse causality, as the decisions are made primarily on the basis of national defense considerations and are therefore not driven by local economic conditions.

This process, known as Base Realignment and Closure (BRAC), is employed by the DoD to either close or "realign" (move soldiers and civilian employees out of or into existing locations) domestic military bases. The closing of military bases has long been a controversial topic, as the loss of jobs – both first-order jobs on a base itself, and second-order jobs in the community that rely on the base and its personnel – raises the concern of constituent communities and their representatives. While both politicians and the military recognized the need to change the military force structure after the end of the Cold War, no politician wanted cuts to happen in their district (Sorenson, 1998). The BRAC process was designed to insulate these decisions from lobbying, political pressure, and conflict between the executive and legislative branches, keeping the focus on military effectiveness and cost efficiency (Mann, 2019). Despite a relatively stream-lined process from a military perspective, however, the affected communities face significant uncertainty about their futures. The BRAC process affects many thousands of jobs, and can result in both net job gains or net job losses. BRAC would often close entire facilities completely. Bases could be listed in the initial proposal and then removed entirely by the end, or appear minimally affected in the initial stages only to face extensive losses in the final results. We discuss BRAC in more detail in Section 2.

Our identification strategy rests on four features:

1. Recommendations are made primarily based on military readiness and military cost savings, and not the affected local economy;
2. The impact of recommendations can be quite large, and may be positive or negative for the affected community;
3. A substantial window of uncertainty exists during which communities and their officials lobby to save jobs at their bases, or gain even more, with occasional success; and
4. The actual change in jobs (the first-moment level effect) happens far removed - often years - from the second-moment uncertainty effect.

Our main results can be summarized as follows. We find that uncertainty generated by BRAC rounds has economically meaningful effects on local economies. Metropolitan areas with greater exposure to BRAC-related uncertainty experience significant declines in the growth of both employment and labor force participation, while exhibiting little change in the unemployment rate. We also find that uncertainty reduces new housing construction, consistent with weaker local housing demand during periods of heightened uncertainty. To identify these effects, we use local attention to military base closures during national BRAC rounds as an instrument for the intensity of the uncertainty shock. The estimated effects suggest that uncertainty depresses economic activity even in the absence of contemporaneous first-moment shocks, providing causal evidence on a central mechanism emphasized by Bloom (2009). Relative to estimates from aggregate time-series models, the effects emerge earlier and are somewhat smaller in magnitude.

Recent research has increasingly focused on identifying the causal effects of uncertainty shocks. Ludvigson et al. (2021) propose a novel structural VAR identification strategy to isolate exogenous uncertainty shocks. Baker et al. (2024) use natural disasters, terrorist attacks, and political shocks as instruments for first- and second-moment shocks, identified using stock returns and volatility. Coibion et al. (2024) and Kumar et al. (2023) employ randomized information treatments that provide different types of information about the first and/or second moments of future economic growth to generate exogenous changes in households' and firms' perceived macroeconomic uncertainty. Our study differs from this literature in two important respects. First, we exploit a natural experiment involving domestic U.S. military base closures, which provides a setting in which uncertainty can be separated from potential first-moment shocks. Second, rather than relying on aggregate time-series variation or survey-based interventions, we identify the effects of uncertainty using geographically localized exposure to BRAC-related uncertainty.

While recent studies have examined the effects of uncertainty at the state and sectoral levels (Shoag and Veuger, 2016; Mumtaz et al., 2018; Baker et al., 2022; Choi et al., 2024), less is known about its effects on local labor markets. We address this gap by analyzing uncertainty shocks at the level of Metropolitan and Micropolitan Statistical Areas (collectively referred to as Core Based Statistical Areas, or CBSAs). CBSAs are an ideal unit of disaggregation for our study, as they capture economic linkages between cities and their surrounding areas where most commuters live. This is discussed in more detail in Section 4.1. Furthermore, we add to this literature by exploring its effects on labor force participation and migration-related channels. The literature has generally focused on the unemployment rate, employment, investment, and output. In contrast, our results suggest that when heightened uncertainty remains unresolved for an extended period, workers may adjust through changes in labor force participation or migration decisions, causing uncertainty shocks to be absorbed through labor supply rather than higher unemployment. We also find that new housing construction responds negatively to uncertainty shocks, consistent with theory.

2 The Base Realignment and Closure Process

Historically, the President had wide latitude to close bases with little input from Congress. However, the US military used to be relatively small in peacetime, only seeing large military expenditures during wartime. Military bases also used to be fairly crude, and so their simple structures were often abandoned when no longer necessary or just given back to local communities. After the Korean War, the United States military was no longer effectively demobilized after the end of hostilities. Security needs during the Cold War gave rise to a large standing army in peacetime for the first time in US history. Congress had good reason to suspect that political opposition to the policies of the Johnson and Nixon administrations lead to retaliation from the executive in the form of base closures. As the Vietnam War wound down in the early 1970s, Defense Secretary McNamara unsentimentally closed many bases, and Congress had enough (Sorenson, 2006).

Congressional legislation blocked the president's ability to unilaterally close bases in 1977 for ten years (10 US Code §2687). Since then there have been five BRAC rounds, beginning in 1988 and followed by rounds in 1991, 1993, 1995, and 2005. When arguing (unsuccessfully) for a new BRAC round in 2018, the Department of Defense (2020) stated that the five previous BRAC rounds had resulted in estimated annual military cost savings of $12 billion, underscoring the economic significance of the process.

One of the key innovations to the BRAC process in 1991 and later was the introduction of an independent commission, whose recommendations came months after the initial recommendations from the DoD, and had to be rejected by Congress or the President in their entirety or they passed by default. The goals of the BRAC commission are laid out in the statute that initiates the process. According to the proposal (Department of Defense, 1991), the process gives "priority consideration" to military value. This emphasis helps separate the BRAC process from local economic conditions – a point we will elaborate on in Section 3.

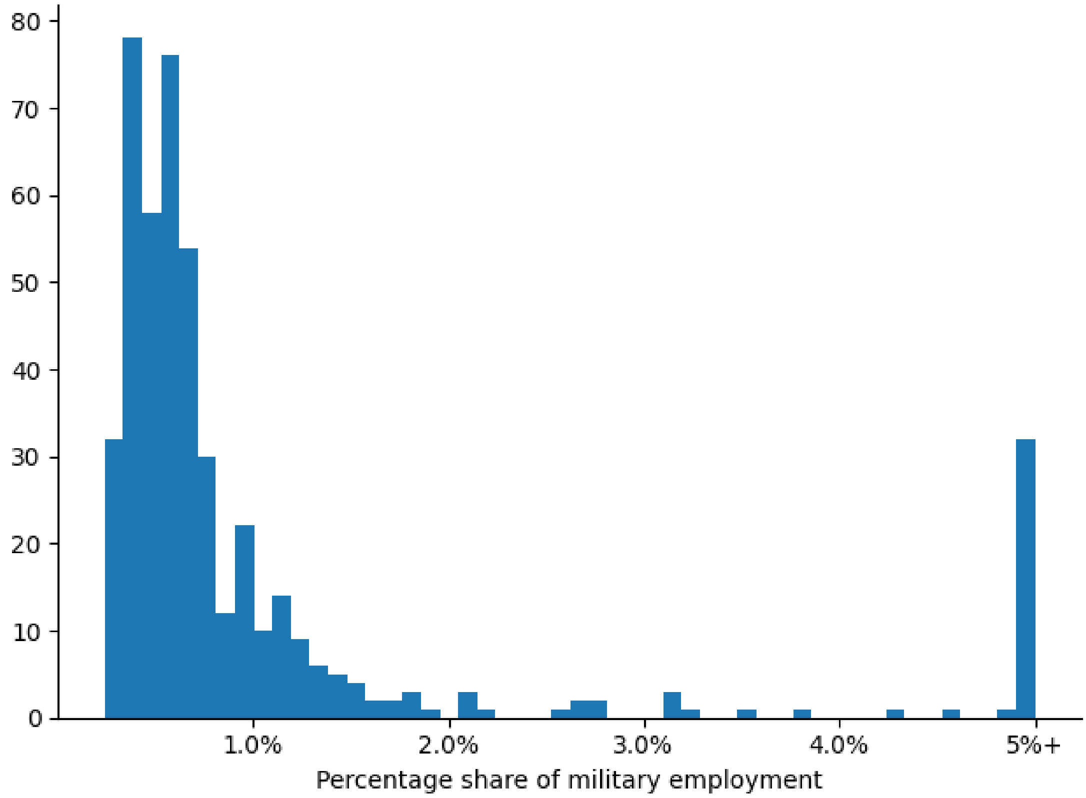
We focus on the 2005 round for several reasons. The 1988 through 1995 rounds were primarily about a reduction in forces following the end of the Cold War, and so there was a significant expectation of general cuts across bases on net. The 1988 round had no public uncertainty period between proposed closures and final results, while the 1990s rounds were all authorized at once and occurred with only a single year between them, making the closures and realignments much more likely to be anticipated, and thus not a clear cause of uncertainty with a discrete start and end. Additionally, closures from the previous rounds were being implemented during the subsequent rounds due to the short gaps, making separation of first and second moment effects challenging. Sorenson (1998) offers further evidence that the 1990s rounds involved a lot of anticipation ahead of time, stating that 1995 was expected to be a round with many cuts, as previous cuts from 1991 and 1993 were "behind schedule."

The 2005 round was different, as it was far removed from previous rounds, and was primarily about a reorientation of military priorities in the context of the War on Terror during a time of rising military expenditures. While some areas experienced cuts, many saw significant gains, and there was no expectation of large negative average changes at any given base. This contrasts with most uncertainty shocks - such as financial crises or natural disasters - which typically involve concurrent negative first-moment effects. In this case, we can focus on the pure second-moment effect due to the temporal lag in implementation and net-zero expectation. We can also control for any potential anticipation of first-moment shocks at the CBSA level, effectively analyzing the impact in the context of a mean-preserving spread. Moreover, the 2005 round was the only BRAC round to affect all 50 states and Washington DC, and data availability is higher in 2005 compared to the rounds in the 1990s, due to its more recent implementation and improved record-keeping. We show the shares of military employment by CBSA metros in 2005 in Figure 1.

The principal steps and dates of the 2005 BRAC process are outlined in Figure 2. The internal processes that preceded the release of the DoD proposal in May included the creation of a force structure report, certification of figures by the Government Accountability Office, and the establishment of the BRAC commission, with members appointed by the president and confirmed by the Senate. Once the BRAC commission makes its report in September, the President can approve the report or return it to the commission, a process that has historically taken less than two weeks, and as little as one day. Congress then has 45 days to reject the entire report, else it is automatically approved. Both Presidential approval and Congressional expiration are generally regarded as formalities that, historically, do not change the final results.[1] The final step, implementation, begins six months later at the earliest, and is required to begin within two years of Congressional

[1]This was an intentional inclusion, by those that set up the BRAC process, to minimize political interference.

Figure 1: Distribution of CBSA Military Employment



Notes: Military share of employment in CBSAs in 2005 was distributed primarily between 0% and 2%, with a small number of outliers beyond the 5% shown (5.4% of our sample CBSAs), extending as far as 37% (for Fort Leonard Wood, MO).

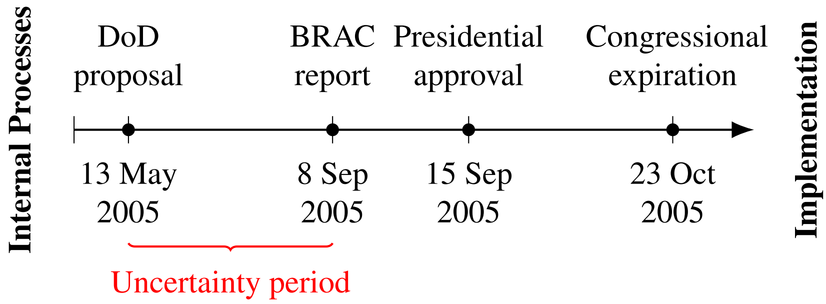
expiration and be completed within six.

3 BRAC as an Uncertainty Shock

We define the BRAC uncertainty period as the time between the Department of Defense's announcement of the initial proposal and the BRAC Commission's final recommendation to the President – that is, from May 13, 2005, to September 8, 2005, as shown in Figure 2.

To identify BRAC as an uncertainty shock, we need to establish five key points: (1) BRAC involves uncertainty, (2) it is exogenous to local economic conditions, (3) it constitutes a shock, (4) it is distinct from a first-moment (level) shock, and (5) no other events that occur concurrently with BRAC and that we cannot control for plausibly explain our effects. We now address each of

Figure 2: Timeline of the 2005 BRAC process



these points in turn.

3.1 BRAC Involves Uncertainty

BRAC recommendations can have positive, negative, or neutral outcomes, and these outcomes shift significantly over the course of the process. In 2005, 13% of the 471 CBSAs in our sample were projected to have employment gains in the initial DoD report, while 23% were projected to have losses. Of those, 21% of them experienced changes in direct military employment outcomes in the time between the initial DoD report and the final BRAC report. These changes during the uncertainty period could involve thousands of jobs, and were almost evenly split between gains and losses, as shown in Table 1. Spending on the BRAC process also had a wide range of outcomes, with more than 10% experiencing gains and more than 30% losses over these four months. In total for the CBSAs in our sample, 36,540 jobs were gained relative to the original DoD report in , while 26,701 jobs were lost.

The final decision on a given base is unknown until the end, making the process inherently uncertain. This uncertainty is heightened by the four-month-long period of public deliberation, during which communities actively lobby to protect their local bases and the associated jobs. According to the final report from the BRAC commission, they "conducted 20 regional hearings to obtain public input and 20 deliberative hearings for input on, or discussion of, policy issues... Commissioners participated in hundreds of meetings with public officials and received well over 200,000 pieces of mail... the Commission's web site registered over 25 million hits." The evolving nature of the recommendations and the very public political and procedural dynamics involved

Table 1: Share of CBSAs with Changes Between DoD Proposal and BRAC Report

	Direct Employment	Spending
Major gain (greater than 25%)	6.8%	6.4%
Minor gain (0 to 25%)	2.1%	4.3%
No change	79.0%	55.3%
Minor loss (-25% to 0)	5.6%	12.8%
Major loss (less than -25%)	6.5%	21.3%

Notes: This table shows the actual changes that occurred during the months between the initial DoD proposal on May 13 and the final BRAC report on September 8. Spending changes without employment changes often represent adjustments to closure and realignment timelines, the chartering of additional studies, or the consolidation of projects and facilities. Some areas with no net change still experienced offsetting changes.

contributed to a widespread perception of uncertainty throughout the BRAC process.

3.2 BRAC is Exogenous to Local Economic Conditions

BRAC decisions are largely exogenous to local economic conditions, based on the criteria established in the statutes that govern the process. Specifically, the criteria for closing or realigning a base are outlined as follows, in order of priority:

Military Value

1. Current and future mission requirements
2. Availability and condition of land
3. Accommodation of future mobilization and contingencies
4. Cost and manpower

Return on Investment

5. Extent and timing of potential costs and savings

Impacts

6. Economic impact on communities
7. Ability of communities to support operations and personnel
8. Environmental impact

The criteria prioritize military value and return on investment, with economic impact on local communities appearing only as the sixth consideration, well after strategic and cost-based factors. The primacy of military goals in the selection process was explicitly stated in the BRAC report (2005): "The Army used the BRAC Selection Criteria during its analyses and ensured that military value (Criteria 1-4) was the primary consideration in making its BRAC 2005 recommendations." Every base affected by the BRAC process is given a section in the final report, with a subsection labeled "Community Concerns," followed by "Commission Findings," which provides further insight into the primacy of military needs over economic impacts.

Of the bases that received community comments during the BRAC uncertainty period, 51% of them included complaints that the DoD did not properly consider the local economic impacts, a number which likely heavily understates the true level of concern due to the perception that the commission would not be swayed by arguments not based on military criteria. Accordingly, complaints about military criteria were included in 100% of the submitted comments from the communities. Of those 51% of bases that also included economic-related comments, the BRAC commission found that the initial DoD report "deviated substantially from final selection criterion 6 [local economic impacts]" in only 12.7% of them. Only 8% of them received a changed outcome, while the remaining received some form of urging from the commission for the DoD to be mindful of the local economy while closing the base.[2] At no point did the BRAC commission ever report implementing a deviation from the DoD recommendations based on the local economy, without that community first fighting for it in the official commenting process, and including a military criteria along with it.

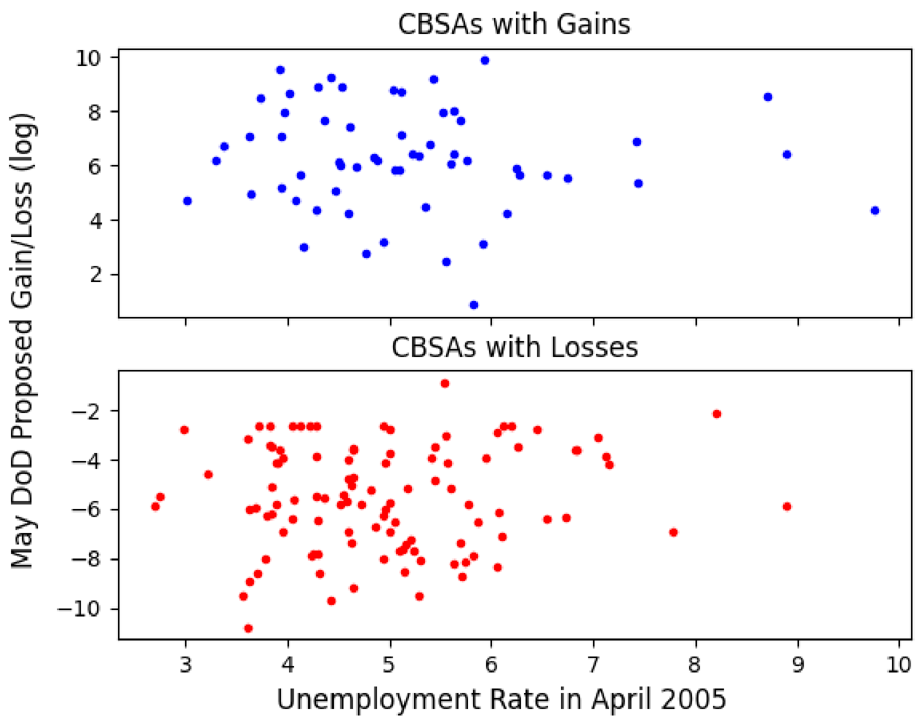
While there is evidence, from policy, official reports, and anecdotal accounts, that local economic conditions were not an important consideration in BRAC selection, we examine this question further by analyzing unemployment rates in CBSAs prior to the start of the BRAC process.

[2]For example, the "Community Concerns" section for Fort Gillem, Georgia, noted severe local economic challenges, including high unemployment and low income. The "Commission Findings" acknowledged flaws in DoD's impact analysis, citing job losses and the base's proximity to a HUBZone. Despite urging mitigation efforts, the Commission proceeded with closure, resulting in 1,081 job losses. Years later, the site remained under EPA Superfund negotiations.

In Figure 3, we present the distribution of unemployment rates in April 2005 for CBSAs that experienced a direct net job change proposed by the DoD. The figure shows no meaningful relationship across these groups. The absence of any clear relationship supports the view that the BRAC process did not place significant weight on local economic conditions when making closure and realignment decisions.

We further formalize this analysis by developing testable hypotheses based on the expected ordering of groups under the assumption that the DoD did take local economic conditions into account. Despite multiple tests (see the appendix for details), we find no evidence that CBSAs selected for losses were economically better off prior to BRAC, nor that CBSAs selected for gains were worse off.

Figure 3: Pre-BRAC CBSA Unemployment Rate by DoD Proposed Job Changes



Notes: Each point represents a single CBSA, showing the relationship between the unemployment rate before the BRAC process began (April 2005), and the inverse hyperbolic sine (to approximate logs for negative values) of the initial DoD report's net employment changes.

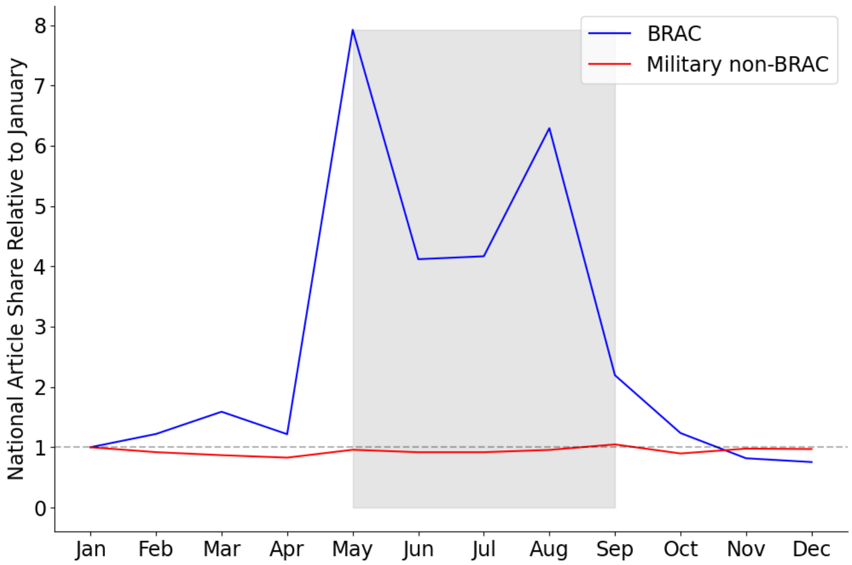
3.3 BRAC Constitutes a Shock

One potential concern is that the BRAC process may either go unnoticed by the public – thus generating little uncertainty – or, conversely, become a constant topic of discussion before the process begins, making the uncertainty period of our natural experiment ill-defined.

To address this issue, we construct a measure of community attention paid to BRAC during the uncertainty period based on the frequency of newspaper coverage. We use the database of U.S. newspapers from Access World News, a well-established source in the macroeconomic uncertainty literature (Baker et al. 2016, Baker et al. 2022). Specifically, we search the archive for articles that mention the Boolean terms *((base and realignment and closure) or brac) and (military or army or navy or marines or "air force").* We then divide those article counts by the total article counts for that month and place.

Figure 4 illustrates the "surprise" nature of BRAC, as national aggregate discussion of BRAC in the news remained relatively flat in the first four months of the year, before jumping by a factor of 8 in May (the month the DoD released its initial proposals), then remaining elevated throughout the BRAC period. The partial drop in September can be explained by the fact that the BRAC process was resolved in the first two weeks of that month, and that some informal results began trickling out at the end of August, as the BRAC commission wrapped up public hearings. To ensure we are capturing attention to BRAC specifically – and not to other concurrent military issues – we also show that discussion of non-BRAC military terms remains flat throughout the year.

Figure 4: National Share of Articles Discussing BRAC and Military Terms in 2005



Notes: The share of U.S. articles in the AWN database from 2005. The BRAC search terms are: *(military or army or navy or marines or "air force") and ((base and realignment and closure) or brac)*. The Military non-BRAC terms are: *(military or army or navy or marines or "air force") not ((base and realignment and closure) or brac)*. Both are divided by the total articles in that month, then normalized to 1 in January. The shaded area represents the BRAC uncertainty period between May and September, inclusive, beginning with the proposals from the DoD and ending with the BRAC report.

To further investigate the unpredictability of BRAC outcomes, we also approach this as a classification problem, considering whether a given CBSA will experience any direct employment changes due to BRAC. We train a logit regression on data for the 1991, 1993, and 1995 rounds. The estimates from these rounds are then applied to data from the 2005 BRAC round to generate the predicted results. Finally, we compare the predicted results for 2005 to the actual outcomes from that year. This yields a "true negative" rate of 49.5% and a "true positive" rate of 9.8%, resulting in an overall prediction accuracy of only 59.3%. Notably, the model exhibits a high "false negative" rate of 39.2%, meaning that in nearly 4 out of 10 cases where a CBSA was actually affected by BRAC-related employment changes, the model incorrectly predicted no impact.[3]

[3]See the Appendix for the details on the model specification.

Finally, when attempting to predict uncertainty across CBSAs during BRAC using structural components (size, unemployment rate, and military share of employment), we find that these factors explain less than 3% of the variation, highlighting the largely unpredictable nature of BRAC-related uncertainty.

3.4 BRAC is Distinct from a First-moment Shock

BRAC represents a second-moment, or uncertainty, shock rather than a first-moment level shock. The key feature that distinguishes BRAC from a traditional first-moment shock is the multi-year delay between the decision and the actual implementation of base closures. Although the economic consequences of BRAC can be significant, implementation typically begins in the fiscal year after the BRAC decision and continues for up to six years. This temporal lag is essential for identifying the effects of uncertainty separately from immediate economic impacts.

3.5 Confounding Events Do Not Obscure BRAC Effects

Many drivers of uncertainty have been identified in the literature, including pandemic lockdown severity (Baker et al., 2022), elections (Baker et al., 2016; Ahir et al., 2022), and natural disasters (Baker et al., 2024).

The U.S. holds major elections in November of even-numbered years, so 2005 saw only a small number of elections, none of which coincided with the BRAC uncertainty period. However, Hurricane Katrina was identified in the Gulf of Mexico on August 23 and made landfall on August 29, 2005. Its effects were large but were isolated to only a small number of CBSAs in our sample along the Gulf Coast region. To control for this event, we run our model with a control for emergency spending and with the affected CBSAs removed, both of which yield similar results. This is discussed in more detail in the following sections.

To rule out other structural causes that may coincide with the timing of the BRAC process, we also run a placebo test using the year 2004. An ideal year for a placebo test should be close to 2005 so that economic conditions are similar, use the same months to account for seasonal effects, and precede the BRAC year so that we are not capturing the delayed first-moment effects of base closures and realignments. While it was known by 2004 that a new round was coming, there was no information on where closures or realignments might occur, nor on the priorities of the DoD for a future round. A national search of BRAC-related articles from the summer of 2004 shows fewer than 10% as many as during the same period in 2005. When we re-run our model (described in Section 5) using all outcome data from the placebo year, we find no statistically significant effects for labor market or housing variables.

Summary of the Identification Strategy

To summarize, the BRAC process constitutes a well-defined uncertainty shock. First, its outcomes are highly uncertain: base realignment and closure recommendations can be positive, negative, or neutral for a given locality, and they are both perceived by the public as changeable throughout the process, and do in fact change for many CBSAs. Following a months-long period of deliberation and lobbying, communities are left uncertain about the final outcome until the BRAC Commission submits its recommendations. Second, BRAC decisions are exogenous to local economic conditions, driven primarily by military value and cost-effectiveness, with local economic impact ranking low in the decision criteria. Third, BRAC is a genuine shock rather than a predictable event. Analysis of U.S. newspaper coverage shows a sharp spike in attention only after the Department of Defense announces its initial proposal, with little coverage beforehand, suggesting that the results of the process were largely unanticipated by the public. Fourth, BRAC is distinct from a first-moment (level) shock. Implementation of closures and realignments is delayed by several years, separating the announcement from its economic consequences. Finally, no substantial, wide-spread confounding uncertainty events that we cannot control for interfere during the BRAC window .

4 Data

4.1 Geographic Level of Analysis

All analysis is performed at the level of Metropolitan or Micropolitan Statistical Areas (collectively referred to as Core Based Statistical Areas, or CBSAs). Each CBSA is a city core plus its surrounding areas. These are an ideal level of disaggregation for this study, since they emphasize economic and social linkages between cities and their surrounding areas. State level analysis would either neglect important economic interactions that cross state lines (as in the New York-Newark-Jersey City, NY-NJ-PA CBSA, or the Memphis, TN-MS-AR CBSA), or questionably assume interactions between very different economic areas in the state (as in the Chicago–Naperville–Elgin, IL–IN–WI CBSA, and parts of southern Illinois). The nature of CBSAs as units of economic connectedness also implicitly controls for spillover effects in our analysis. While it may be justifiable to consider spillover effects – particularly between two CBSAs within the same state where state-level policy actions could be influenced by the BRAC in the long term – the assumption of no spillover effects is reasonable over the short time horizons we are studying.

4.2 Sources

The data on the BRAC round, including both proposed and final base closures and realignments, come from the 1991, 1993, 1995, and 2005 Department of Defense reports to the BRAC commission, and the BRAC commission reports to the president. Military share of employment in a given CBSA comes from the Bureau of Economic Analysis (BEA) regional data.

The CBSA-level employment, labor force, and unemployment rate variables are part of the Bureau of Labor and Statistics (BLS) Local Area Unemployment Statistics series. This employment data is constructed by the BLS based on the Current Population Survey, American Community Survey, annual population estimates, and state unemployment insurance data, then seasonally adjusted. Controls for the impacts of Hurricane Katrina on Gulf-coast states are collected from the Federal Emergency Management Agency (FEMA) Individual and Household Programs, total claims paid data.

Data on the share of new housing loans come from the CoreLogic Loan Level dataset. This large dataset contains more than 40 million individual loans between 1999 and 2011. Our measure is constructed by dividing the number of loans originated for new home construction by the total number of loans originated in a given CBSA and month. The resulting series is then seasonally adjusted.

The news-based measure of attention to BRAC is constructed by searching the Access World News (AWN) database as described in Section 3.3. Our uncertainty measure is constructed similarly, by searching AWN for the terms *(economic or economy or economies) and (uncertain or uncertainty or uncertainties)*. In all news-based measures, the results for each time period and CBSA are divided by the total articles for that time and place, then normalized to have a unit standard deviation.

By contrast, existing news-based measures often focus on policy-related uncertainty, while we focus on more general economic uncertainty for two reasons. One, excluding policy-specific search terms broadens the number of articles we capture, which is particularly important for disaggregated geographies like CBSAs. Two, we have no reason to exclude economic uncertainty that is not explicitly related to policy.

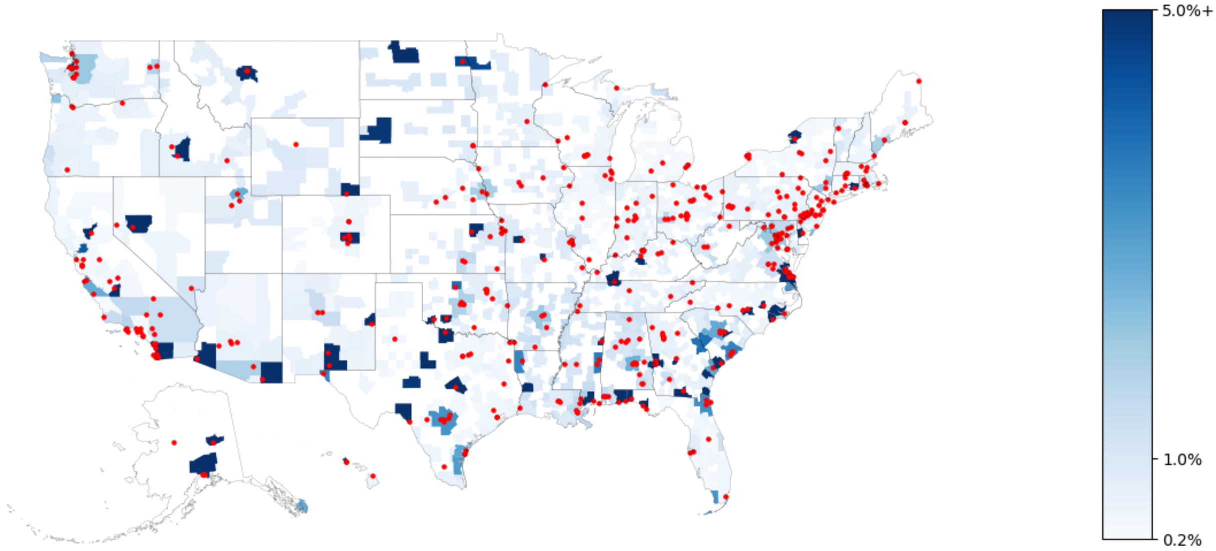
4.3 CBSA Coverage and Military Base Distribution

Our analysis covers 471 CBSAs across all 50 states and Washington DC. This represents 49.5% of all of the CBSAs in the United States in 2005, with our primary constraint being data availability, particularly in the AWN database. All of the CBSAs in our sample have non-zero military employment, with a mean of 1.53% and median of 0.61%. This is very close to the average military share of employment for all CBSAs in the US (1.44% mean and 0.61% median). The full distribution of our sample is in Figure 1. The wide scope of the 2005 BRAC round is illustrated in Figure 5, which shows the bases listed in the May 2005 DoD report distributed across the country. Of the CBSAs in our sample, 167 (35.5%) are home to at least one military base that is listed in the May 2005 DoD report.

5 Estimating the Causal Impact of Uncertainty

This section begins by detailing our analytical framework. We then assess the strength of our instrumental variable before presenting results on how uncertainty affects the labor market and housing market at the CBSA level.

Figure 5: CBSA Military Share of Employment and BRAC Military Base Locations



Notes: Coloration represents the percent of military employment in a given CBSA in 2005, with no color meaning zero military employment, and dark blue meaning 5% or more, with an average slightly over 1%. About 5.4% of CBSAs have 5% or more military employment, which we clip to 5% here in order to focus on the majority of the distribution. Each red dot is the (approximate) location of a military base that was in the May 2005 Department of Defense report.

5.1 Analytical Framework

We aim to estimate the impact of uncertainty on changes in local labor markets from the month before BRAC began to the current month. Specifically, we estimate the following OLS regression:

$$Y_{jt} = \alpha_j + \beta_1 U_{jt} + \beta_2 D_{jt} + \beta_3 F_{jt} + \epsilon_{jt} \tag{1}$$

The dependent variable, Y_{jt}, is either a labor market indicator - the change in the logarithm of employment level, the logarithm of labor force level, or the unemployment rate - or a housing market indicator - the share of loans for new housing construction or the logarithm of average sale price - in CBSA j at time t from the month before BRAC began ($t = 0$): $Y_{jt} = \text{outcome}_{jt} - \text{outcome}_{j0}$. We use the inverse hyperbolic sine transformation in place of the logarithmic transformation, as it closely approximates log results while accommodating both zero and negative values.

Our primary variable of interest, U_{jt}, captures cumulative economic uncertainty in CBSA j at time t, up to the conclusion of the BRAC round at month $t = 5$. After this point, the same value is carried forward. Specifically, U_{jt} is defined as

$$U_{jt} = \begin{cases} \frac{\sum_{i=1}^{t} \text{econ\_unc\_articles}_{ji}}{\sum_{i=1}^{t} \text{all\_articles}_{ji}} & \text{if } t \leq 5 \\ \frac{\sum_{i=1}^{5} \text{econ\_unc\_articles}_{ji}}{\sum_{i=1}^{5} \text{all\_articles}_{ji}} & \text{if } t > 5 \end{cases}$$

For ease of interpretation, we normalize U_{jt} to have a unit standard deviation. This construction allows us to isolate the effect of uncertainty specifically generated during the BRAC process, excluding any uncertainty arising in the post-BRAC period.

The control variable, D_{jt}, captures the direct impact of BRAC in CBSA j. Since no actual level changes due to BRAC occur within our time frame, D_{jt} serves to control for anticipation effects. For all $t \leq 5$ (i.e., before the BRAC round concludes), the direct effect is measured as the log of the projected employment changes reported in the May 2005 DoD report. For all $t > 5$ (i.e., after the conclusion of the round in September), it is measured as the log of the employment effects based in the final BRAC Commission report. In particular,

$$D_{jt} = \begin{cases} \text{DoD\_proposed}_j & \text{if } t \leq 5 \\ \text{BRAC\_final}_j & \text{if } t > 5 \end{cases}$$

Finally, F_{jt} is the log of total FEMA claims in Gulf Coast states from September onward ($t \geq 5$), included to control for the effects of Hurricane Katrina:

$$F_{jt} = \begin{cases} 0 & \text{if } t < 5 \\ \text{katrina\_claims}_j & \text{if } t \geq 5 \end{cases}$$

For each of the five dependent variables, Eq. (1) is estimated 12 times for each $t \in [1...12]$, resulting in a total of 60 regressions. However, these regressions may be biased due to potential endogeneity of the economic uncertainty variable, that is, $\text{corr}(U_{jt}, \epsilon_{jt}) \neq 0$.

To address these biases, we propose instrumenting economic uncertainty using attention paid to the BRAC process. This instrument allows different CBSAs to experience heterogeneous levels of uncertainty in response to BRAC, under the defensible assumption that CBSAs that are more concerned with the effects of BRAC for any reason will discuss it more, while CBSAs with any information that makes them less concerned with the BRAC process will discuss it less. Specifically, our first-stage regression is:

$$U_{jt} = \alpha_j + \gamma_1 B_{jt} + \gamma_2 D_{jt} + \gamma_3 F_{jt} + \delta_{jt} \tag{2}$$

where B_{jt} is the news-based measure of the number of articles referencing BRAC during the process, and the other variables are as defined in Eq. (1). The variable B_{jt} captures cumulative attention in CBSA j at time t, up to the conclusion of the BRAC round at month $t = 5$. After this point, the same value is carried forward. Specifically, B_{jt} is constructed as:

$$B_{jt} = \begin{cases} \frac{\sum_{i=1}^{t} \text{brac\_articles}_{ji}}{\sum_{i=1}^{t} \text{all\_articles}_{ji}} & \text{if } t \leq 5 \\ \frac{\sum_{i=1}^{5} \text{brac\_articles}_{ji}}{\sum_{i=1}^{5} \text{all\_articles}_{ji}} & \text{if } t > 5 \end{cases}$$

We then proceed with the second-stage regression:

$$Y_{jt} = \alpha_j + \beta_1 \hat{U}_{jt} + \beta_2 D_{jt} + \beta_3 F_{jt} + e_{jt} \tag{3}$$

where $\hat{U}_{jt}$ is the predicted values from Eq. (2).

Our instrumental-variable strategy relies on the assumption that local attention to BRAC influences economic outcomes through uncertainty rather than through any direct economic channel. This assumption is supported by the institutional features of the BRAC process. During the uncertainty period, no actual base closures, realignments, or spending changes occur, and implementation typically begins years later. Moreover, we explicitly control for both the proposed and final BRAC employment effects to account for anticipation of future changes. Under these conditions, differences in BRAC-related news coverage across CBSAs primarily capture variation in the intensity of uncertainty generated by the deliberation process itself. As a result, attention to BRAC provides a plausible source of exogenous variation in local uncertainty.

Eq. (3) provides unbiased estimates, conditional on B_{jt} in Eq. (2) being a strong instrument, a condition we test and demonstrate in the following section.

5.2 Attention to BRAC is a Strong Instrumental Variable

The commonly accepted metric for instrument strength is the F-statistic from the first-stage regression in Eq. (2). In our exactly-identified model with one endogenous variable and one instrument, the first-stage F-statistic is equivalent to the square of the t-statistic on the instrument. For the five months of the BRAC process, the F-statistics are 14.7, 30.6, 23.3, 20.0, and 18.4, respectively. After month 5, when the BRAC process concludes, the news-based measures are no longer updated, yielding a constant F-statistic of 19.4 for the post-BRAC period. All first-stage F-statistics exceed the rule-of-thumb threshold of 10 suggested by Staiger and Stock (1997).[4]

More recently, Olea and Pflueger (2013) develop a test for weak instruments that is robust to heteroskedasticity, autocorrelation and clustering. Our estimates exceed the critical values of this more stringent test at the 5% significance level for a 10% maximum bias in months 2 and 3, only narrowly missing this threshold in months 4-5. In those months, the statistics exceed the critical values corresponding to a maximum 20% bias relative to OLS.[5]

Aside from directly testing the strength of instruments, we also use estimation methods that are robust to weak instrumental variables. Moreira (2009) and McElroy and Sheng (2026) argue that, in the case of a single instrument, confidence intervals from Anderson and Rubin (1949) are efficient regardless of whether instruments are strong or weak. We calculate Anderson-Rubin confidence intervals and show them alongside OLS intervals in Figure 6. In nearly all cases, they are shifted further away from zero than standard OLS confidence intervals.

[4]While first-stage results are not a function of the second-stage outcome variable, the F-statistics vary slightly for our housing loan model due to a lack of loan data for a small number of CBSAs that differ in each period (while the labor market data has the same CBSA matched in every period.) They are generally slightly smaller but very similar in all periods except month 6. See Table 2 and Table 3.

[5]Olea and Pflueger critical values are: max 10% bias = 23.1, max 20% bias = 15.1, max 30% bias = 12.

Table 2: Effects of Uncertainty From the 2005 Base Realignment and Closure on Local Labor Market

Dependent Variable	Equation	Months Since the Beginning of BRAC Round											
		BRAC Uncertainty Period					Post-BRAC Period						
		1	2	3	4	5	6	7	8	9	10	11	12
employment	OLS	0.06*	0.07	0.02	0.01	0.10	0.13	0.12	0.11	0.15	0.16	0.16	0.09
		(0.04)	*(0.06)*	*(0.07)*	*(0.08)*	*(0.09)*	*(0.10)*	*(0.09)*	*(0.10)*	*(0.10)*	*(0.11)*	*(0.11)*	*(0.11)*
labor_force	OLS	-0.07**	0.05	-0.01	0.04	0.13	0.01	0.12	0.12	0.16*	0.15	0.16	0.08
		(0.03)	*(0.05)*	*(0.07)*	*(0.07)*	*(0.09)*	*(0.09)*	*(0.09)*	*(0.08)*	*(0.09)*	*(0.10)*	*(0.11)*	*(0.11)*
unemp_rate	OLS	-0.01	-0.00	-0.02	0.02	-0.02	-0.01	-0.01	-0.01	0.01	0.01	-0.01	-0.02
		(0.01)	*(0.01)*	*(0.02)*	*(0.02)*	*(0.05)*	*(0.05)*	*(0.05)*	*(0.03)*	*(0.03)*	*(0.03)*	*(0.03)*	*(0.03)*
employment	2SLS	-0.04	-0.51**	-0.96**	-0.90**	-1.05*	-0.66	-0.67	-0.50	-0.44	-0.28	-0.26	-0.50
		(0.20)	*(0.24)*	*(0.40)*	*(0.43)*	*(0.55)*	*(0.54)*	*(0.50)*	*(0.50)*	*(0.52)*	*(0.56)*	*(0.56)*	*(0.58)*
labor_force	2SLS	-0.04	-0.50**	-0.88**	-0.82**	-1.02**	-0.71	-0.73	-0.47	-0.38	-0.34	-0.29	-0.51
		(0.18)	*(0.23)*	*(0.36)*	*(0.39)*	*(0.52)*	*(0.50)*	*(0.48)*	*(0.44)*	*(0.48)*	*(0.53)*	*(0.54)*	*(0.55)*
unemp_rate	2SLS	-0.01	0.02	0.06	0.02	-0.04	-0.12	-0.10	-0.00	0.04	-0.02	-0.02	-0.03
		(0.06)	*(0.06)*	*(0.09)*	*(0.10)*	*(0.27)*	*(0.25)*	*(0.24)*	*(0.16)*	*(0.14)*	*(0.14)*	*(0.14)*	*(0.14)*
First-Stage F-Stat		14.7	30.6	23.3	20.0	18.4				19.4			
Note:	*p<0.1; **p<0.05; ***p<0.01												

Notes: The results of six different specifications over 12 time periods (72 regressions) are listed above. The first three rows give the coefficient estimates from the OLS regression (Eq. 1) with three different dependent variables. The second three rows give the coefficient estimates from the IV regression, when instrumenting uncertainty with a measure of BRAC mentions in the news (Eq. 3). The first stage regression is the same for every labor market outcome in the post-BRAC period because any uncertainty after the resolution of the BRAC process would no longer be explainable by our instrument, so we carry the values forward after month five. The employment and labor force values are in 100*log changes from the time period before BRAC, while the unemployment rate is in percentage point changes. The uncertainty variable is normalized to unit standard deviation.

Table 3: Effects of Uncertainty From the 2005 Base Realignment and Closure on Local Housing Market

Dependent Variable	Equation	Months Since the Beginning of BRAC Round: BRAC Uncertainty Period					Post-BRAC Period						
		1	2	3	4	5	6	7	8	9	10	11	12
new_const_loans	OLS	-0.14*	-0.09	-0.13*	0.12	-0.03	-0.03	-0.07	-0.06	0.02	0.03	-0.11	-0.05
		(0.08)	*(0.08)*	*(0.06)*	*(0.09)*	*(0.09)*	*(0.09)*	*(0.08)*	*(0.10)*	*(0.08)*	*(0.07)*	*(0.09)*	*(0.09)*
avg_sale_price	OLS	-1.72	4.03	-0.87	2.62	-0.99	0.97	3.13	-0.58	1.47	0.51	1.48	0.17
		(1.49)	*(2.77)*	*(1.50)*	*(1.51)*	*(1.57)*	*(3.01)*	*(3.06)*	*(2.95)*	*(3.01)*	*(3.81)*	*(1.60)*	*(1.43)*
new_const_loans	2SLS	-1.05**	-0.53	-0.85**	-1.13**	-1.38**	-0.64	-1.03**	-1.05*	0.84*	0.73*	-1.16*	-0.48
		(0.52)	*(0.33)*	*(0.34)*	*(0.50)*	*(0.59)*	*(0.94)*	*(0.48)*	*(0.56)*	*(0.44)*	*(0.43)*	*(0.55)*	*(0.49)*
avg_sale_price	2SLS	-3.21	7.29	-4.12	-4.96	0.44	2.29	7.74	0.07	8.17	18.74	4.76	2.00
		(8.51)	*(11.18)*	*(6.91)*	*(7.60)*	*(8.17)*	*(15.21)*	*(15.52)*	*(14.93)*	*(15.29)*	*(19.73)*	*(8.14)*	*(7.24)*
First-Stage F-Stats		14.2	27.8	20.0	17.9	16.2	3.9	16.8	16.1	17.9	17.0	16.7	17.3

Note: *p<0.1; **p<0.05; ***p<0.01

Notes: The results of four different specifications over 12 time periods (48 regressions) are listed above. The first two rows give the coefficient estimates from the OLS regression (Eq. 1) with two different dependent variables. The second two rows give the coefficient estimates from the IV regression, when instrumenting uncertainty with a measure of BRAC mentions in the news (Eq. 3). Unlike in Table 2, the first-stage F-statistics vary in the post-BRAC period for the housing regressions due to slight changes in matched CBSAs over time. Month six is an outlier here due to a drop in matched CBSAs with data during that period. The average sale price variable is in 100*log changes from the time period before BRAC, while the new housing construction loan share is percentage point changes. The uncertainty variable is normalized to unit standard deviation.

5.3 Labor Market Results

Figure 6 presents the impacts of uncertainty on local labor markets based on the two-stage least squares (2SLS) regression in Eq. (3). Table 2 contains the detailed 2SLS estimates, compared to those from the OLS regression in Eq. (1). Instrumenting economic uncertainty with attention to BRAC, we find that a CBSA with one standard deviation higher economic uncertainty experiences an employment level about 0.1 log points lower than it otherwise would have been; see Panel (a).[6] This is similar in magnitude to the peak employment effect found in Baker et al. (2022), where a standard-deviation shock to uncertainty leads to a peak drop in employment of 0.13 log points in a panel VAR framework.

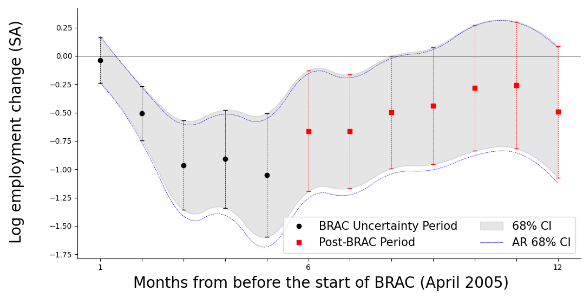
Furthermore, our estimates indicate that the peak effect on employment occurs about five months after the beginning of the BRAC period, which is notably sooner than the peak response of 11–14 months found in Baker et al. (2022). Some of these differences between our results and those from the VAR analysis are undoubtedly due to the characteristics of the specific natural experiment we are using, rather than a more general shock to uncertainty.

Turning to the impact on the unemployment rate in Panel (b), we do not observe any significant results. This finding contrasts with most studies that document large and persistent unemployment effects in response to uncertainty shocks. To explore this further, we examine the labor force response, given that the unemployment rate is a function of both the employment level – which we observe similar results for – and the labor force. To our knowledge, the previous literature has not directly explored the labor force response to uncertainty. However, Baker et al. (2022) finds a peak *unemployment rate* response that, while significant, is about 21% smaller than the peak *employment level* response using state-level data. While the authors do not address this directly, it can be explained by a same-direction response in the labor force of smaller magnitude than the response in employment. Given our focus on the CBSA level, we are able to examine labor market

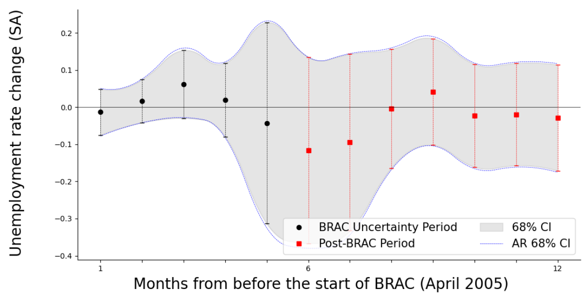
[6]We also explored adding a control for sentiment, by looking at BRAC-releated articles from a smaller sample of 128 CBSAs available in the ProQuest database. Unlike in our larger AWN sample, ProQuest provides programmatic access to the full text of articles through their TDM Studio. We constructed sentiment using the Spacy and textblob Python libraries, which considers not just the score for an individual word, but also negations and amplifiers (e.g. "very bad" < "bad" < "not good" < "good" < "very good"). The sentiment control did not qualitatively change our results.

Figure 6: Impacts of 2005 BRAC Uncertainty Shock on Local Labor Market

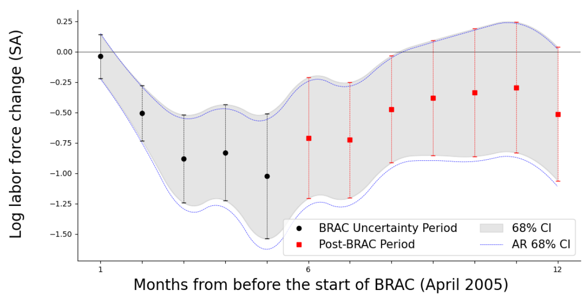
(a) 100 * Log Employment Change, Seasonally Adjusted (SA)



(b) Unemployment Rate Change, Seasonally Adjusted (SA)



(c) 100 * Log Labor Force Change, Seasonally Adjusted (SA)



Notes: Each point shown on these figures represents the coefficient estimate on the economic uncertainty variable from Eq. (3), using attention to BRAC as the instrumental variable. The shaded area represents one standard error confidence intervals, and the blue lines represent Anderson-Rubin confidence intervals, which are robust to weak instruments. The significant and similar magnitude drop in log employment (Panel a) and log labor force (Panel c) account for the insignificant effect on the unemployment rate (Panel b). See Table 2 for detailed values.

dynamics between CBSAs that may be overlooked in studies using more aggregated data.

Our results show an almost identical negative response in the labor force (Panel (c)) to an uncertainty shock as we observe in employment, with the loss peaking at about 0.1 log points in month five. This suggests that the lack of a response in the unemployment rate may be explained by some combination of people moving away from affected CBSAs during the BRAC process, or, more likely, locals ceasing to search for new work, or potential newcomers deciding to move elsewhere due to heightened uncertainty. Unfortunately, we cannot delve into the details of this process, as county-to-county migration flows are estimated by the Census based on American Community Survey data, which are not available at frequencies that align with the natural experiment we are analyzing. However, when considering the aggregate macroecononomy, one can forget the importance of disaggregated flows, which are important for the American labor market (Davis et al., 2012). The United States sees large flows between metro areas, and this type of inter-metro migration is strongly pro-cyclical, even when accounting for local economic conditions (Saks and Wozniak, 2011). It is, then, perhaps not surprising that Americans simply don't move to places where economic activity is being depressed due to uncertainty.

The direct proposed DoD and BRAC effects, which we use to account for anticipation given the information available at the time, are not significant at any time horizon. The FEMA claims, which enter the equations only from September onward, are highly significant. In our sample, 22 CBSAs (about 4.6%) had FEMA claims. Due to the magnitude of the damage caused by Hurricane Katrina, we also re-run the regression with these CBSAs dropped from the sample and find that this does not meaningfully change the sign, magnitude, or significance of any of our results relative to using the full control sample.

While we are cautious in drawing strong conclusions about the persistence of the employment and labor force effects over time, our post-BRAC periods exhibit a pattern broadly consistent with impulse response functions in the VAR literature – showing a gradual return to baseline following the peak. Although the second-stage regression results for the later periods are not statistically significant, this may reflect limited statistical power rather than an absence of underlying effects, particularly given that we do not include the purely endogenous uncertainty that arises after the BRAC natural experiment ends.

5.4 Direct Impact of BRAC

We can also estimate the economic significance of uncertainty which is directly caused by the BRAC process. Focusing on the peak effect between April and September (period 5 in Table 2), we first take the attention to BRAC measure, B_{jt}, then multiply it by its estimated coefficient from the first stage (Eq. 2), and then by the estimated coefficient on economic uncertainty from the second stage (Eq. 3). This value, $\gamma_1 \beta_1 B_{jt}$, is the estimated direct effect of a standard-deviation higher attention to BRAC on the labor market outcome variables in a given CBSA by September 2005, at the end of the BRAC process. Table 4 shows the results at different percentiles of job changes in the initial DoD proposal.

To put this magnitude in perspective, the estimated BRAC effect is roughly one-fifth as large as the decline in nonfarm employment growth during the 1990–91 recession and about one-sixth as large as the decline observed during the 2001 recession (Singleton, 1993; CBO, 2005).[7] The effect is also economically significant when we compare our estimates to the level of military employment. In 2005, military employment accounted for 1.2% of total nonfarm employment.[8] While this BRAC shock affected the entire economy, it was equivalent to eliminating a sixth of the military jobs in a CBSA in 2005 – a significant result given that this is a second, and not a first, moment shock. This is much larger than the first-moment of net job cuts for the 2005 BRAC Round, which was less than a fiftieth of a percent of total nonfarm employment.

[7]Our effect peaks around 5-6 months after the impulse, while the 1990-91 and 2001 recessions both lasted 8 months.

[8]Military employment is from BEA, Series B4280C0A173NBEA. Total nonfarm employment is from the BLS, Current Population Survey.

Table 4: Direct Effects of BRAC Attention on Selected CBSAs

CBSA	DoD Proposal		BRAC Result			Employment		Labor Force	
	Percentile	Level	Level	B_{jt}	$\gamma_1\beta_1 B_{jt}$	Total Δ	BRAC Δ	Total Δ	BRAC Δ
Atlanta, GA	1st	-6,696	-6,772	0.601	-0.123	9,217	-3,013	23,364	-3185
Davenport, IA	10th	-1,263	-1,441	0.903	-0.185	3,132	-356	4,556	-373
Clarksville, TN	20th	-351	-351	0.000	0.000	1,399	0	783	0
Maysville, KY	50th	-18	-18	0.057	-0.012	-83	-1	-154	-1
Oklahoma City, OK	80th	203	-237	0.530	-0.109	2,984	-596	616	-624
Providence, RI	90th	607	861	0.507	-0.104	1,194	-684	3,625	-722
Baltimore, MD	99th	5,297	3,146	0.337	-0.069	13,277	-899	10,771	-941

Notes: This table shows CBSAs at selected percentiles based on the magnitude of job changes in the initial DoD proposal in April 2005, excluding areas that had no proposed changes. The BRAC result shown is the actual change at the end of the uncertainty period, in September 2005, while B_{jt} is the attention to BRAC instrumental variable described in Eq. (2). For each of the employment level and the labor force level, the column $\gamma_1\beta_1 B_{jt}$ is the response of the outcome to a unit standard deviation rise in attention to BRAC; the total change is the observed seasonally-adjusted level change between April and September 2005; and the BRAC change is the level change our model estimates is attributable to attention to BRAC in that CBSA during the uncertainty period. Note that $\gamma_1\beta_1 B_{jt}$ is different between the Employment and Labor Force regressions due to the presence of β_1, but at the level of precision shown the values are the same for all CBSAs in this table, so for the sake of brevity the value is only shown once.

5.5 Housing Market Results

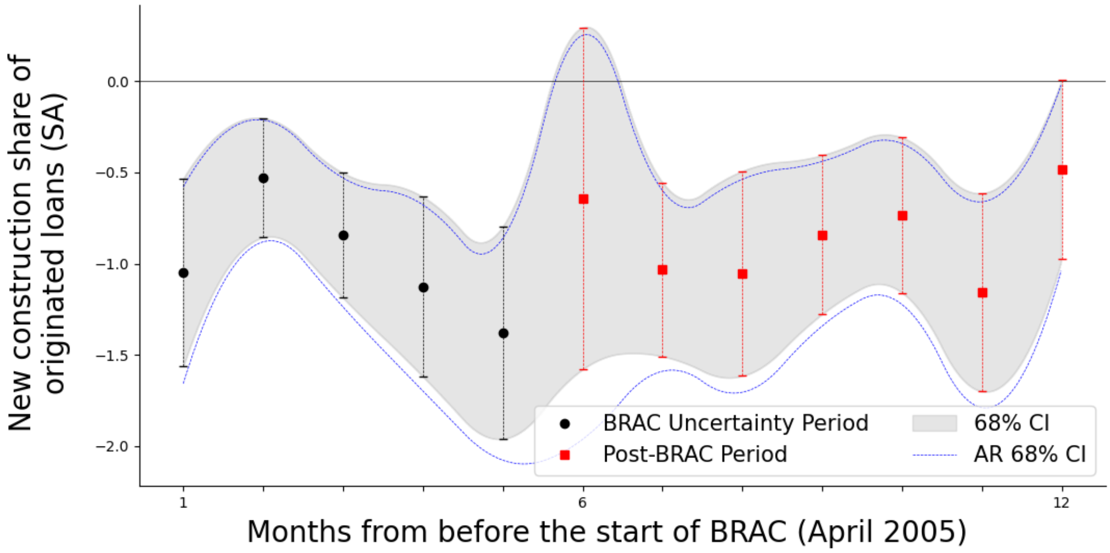
We extend our analysis beyond labor market outcomes to examine the effects of uncertainty on the housing market. See Figure 7 for these results, with details on the OLS and 2SLS estimates in Table 3. Using the large CoreLogic Loan dataset, we are able to access fine-grained data on originated loans in each CBSA over time. There is strong evidence that housing prices are sticky (Guren, 2018; Fan, 2022), as homeowners seeking to sell are reluctant to lower the asking price of what is often their largest asset. Our results are consistent with this finding, without significant effects of uncertainty on the sale or appraised prices of properties during the BRAC period.

Instead, following the models of Bernanke (1983), *inter alia*, that link the impact of an uncertainty shock to the degree of irreversibility of an investment,[9] we look at the share of loans that go toward new construction (as opposed to purchases, refinancing, remodeling, and debt consolidation). New home construction is a lengthy process that requires a substantial time commitment during which there is no completed home to sell, making new home construction significantly less reversible in the short run than the purchase or refinance of an existing home. We find a small but significant 0.14 log point decrease in response to a unit standard deviation increase in uncertainty. The drop occurs immediately and is persistent across our time horizon.[10] The housing market results are also consistent with our findings for employment and labor force participation. The decline in new home construction suggests weaker housing demand in areas experiencing greater BRAC-related uncertainty, a pattern that is consistent with reduced in-migration to affected locations.

[9] See the survey in Pindyck (1991) for a survey of major works in this literature.

[10] We also attempt to validate this result using the publicly-available Census Building Permits Survey. However, that data covers far fewer CBSAs than the CoreLogic data, cutting out over 45% of our observations relative to our models using labor market outcomes. While the results are similar to ones found with the CoreLogic data, they are noisy and insignificant at all time periods.

Figure 7: Impacts of 2005 BRAC Uncertainty Shock on Home Construction Loans



Notes: Each point shown on these figures represents the coefficient estimate on the economic uncertainty variable from Eq. (3), using attention to BRAC as the instrumental variable. The shaded area represents one standard error confidence intervals, and the blue lines represent Anderson-Rubin confidence intervals, which are robust to weak instruments.

6 Conclusion

In this paper, we adopt a novel approach to examining uncertainty by utilizing an exogenous natural experiment to isolate the effects of second-moment uncertainty shocks from any potentially confounding first-moment effects. The BRAC process serves as an ideal means of achieving this identification by separating level effects – often by several years – from uncertainty effects, which allows us to examine the process across hundreds of locations. After accounting for anticipation of the direct effect of changes due to BRAC, we observe significant declines in employment growth, labor force growth, and the share of loans going toward new home construction. This pattern is consistent with weaker local labor and housing demand in areas experiencing greater BRAC-related uncertainty. Additionally, variation in the intensity of BRAC-related uncertainty across CBSAs provides a rich source of identification, enabling us to estimate how local economic outcomes respond to uncertainty shocks of differing magnitudes.

Our two-stage least squares estimates indicate that a CBSA with a one-standard-deviation increase in uncertainty experiences, on average, a 1.05 percentage point decline in employment growth, a 1.02 percentage point decline in labor force growth, and a 1.14 percentage point decline in the share of loans used for new home construction. In the BRAC setting, the average CBSA in our first-stage regression experienced a one-fifth standard deviation increase in uncertainty in response to a one-standard-deviation increase in attention to BRAC during the uncertainty period. In the case of other uncertainty shocks, the source of the disturbance is rarely identified as cleanly, which limits the ability to trace how the initial uncertainty propagates through the economy. Future work examining similar episodes with well-identified uncertainty shocks could further isolate these effects, particularly if their impacts on the labor force can also be measured.

Although the validity of our treatment effect relies, to some extent, on the assumption that policymakers involved in a BRAC round prioritize military and strategic considerations over local economic conditions, our findings suggest that the uncertainty generated by the BRAC process itself has economically meaningful consequences for affected communities. These costs arise well before any actual base closures or realignments occur, indicating that the economic effects of uncertainty should be considered separately from the effects of the underlying policy actions. More broadly, the results underscore the importance of accounting for uncertainty when evaluating large-scale policy changes that involve lengthy periods of public deliberation and review.

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Appendix

DoD Proposals Do Not Reflect Pre-Existing Unemployment Conditions

Here we test whether the expected ordering of CBSAs grouped by gains, no net change, or losses, in the May DoD proposal holds if we assume the DoD considered economic conditions in their process (i.e. that high unemployment CBSAs were more likely to experience gains and low experience losses). First, using an ordered logit, we attempt to classify a CBSA by whether the DoD proposed net gains, losses, or neither based on the April unemployment rate. Second, we use the Jonckheere-Terpstra test to non-parametrically test the hypothesis that the pre-BRAC unemployment rate in places with losses was higher than in untreated places, which was higher than in places with gains. The ordered logit coefficient on the pre-BRAC unemployment rate is positive, but small and statistically insignificant, while the non-parametric test rejects the hypothesis that the expected ordering holds.

An additional possibility we can test for is that the military identifies a need, e.g. consolidation of two bases that perform similar duties, then decides which base gains and which base looses based on economic conditions. To directly test this expected ordering without considering the CBSAs with no net changes, we run a Kolmogorov-Smirnov two-sample test, where we again reject the hypothesis that BRAC gains and losses correspond to the expected ordering of the pre-BRAC unemployment rate.

Prediction of BRAC Results in 2005

The model trained on data from 1991, 1993, and 1995 rounds is specified as follows:

$$\begin{aligned} Y_{jt} = \alpha_j &+ \beta_1 D_{jt} + \beta_2 P_{jt} \\ &+ \beta_3 Unemp_{jt} + \beta_4 Mil_{jt} + \beta_5 Agr_{jt} + \beta_6 Mfg_{jt} \\ &+ \beta_7 Vc_{jt} + \beta_8 Pc_{jt} + \beta_9 Nw_{jt} + \beta_{10} Pd_{jt} + \epsilon_{jt} \end{aligned} \tag{A1}$$

The dependent variable Y_{jt} equals 1 if the CBSA j will experience any direct employment changes from the BRAC in year t, and 0 otherwise. The explanatory variables are

- D_{jt} is the log of the proposed employment changes from the DoD at the beginning of the BRAC process;
- P_{jt} is the log of the direct employment results of the BRAC process from the prior round;
- $Unemp_{jt}$ is the seasonally-adjusted unemployment rate before the start of the BRAC round;
- Mil_{jt}, Agr_{jt} and Mfg_{jt} are the military, agricultural and manufacturing share of employment, respectively;
- Vc_{jt} and Pc_{jt} are the violent crime and property crime rates, respectively;
- Nw_{jt} is the non-white share of the population;
- Pd_{jt} is the population density.

The estimates from the 1991, 1993, and 1995 rounds for the intercept a_j and the slope parameters β_1 through β_{10} are applied to the 2005 BRAC round to generate the new $\hat{Y}_{j,2005}$. We then compare the predicted results for 2005 to the actual results from 2005. This yields a "true negative" rate of 49.5% and a "true positive" rate of 9.8%, resulting in an overall prediction accuracy of only 59.3%. Notably, the model exhibits a high "false negative" rate of 39.2%, indicating that a substantial portion of actual positives were incorrectly classified as negatives.